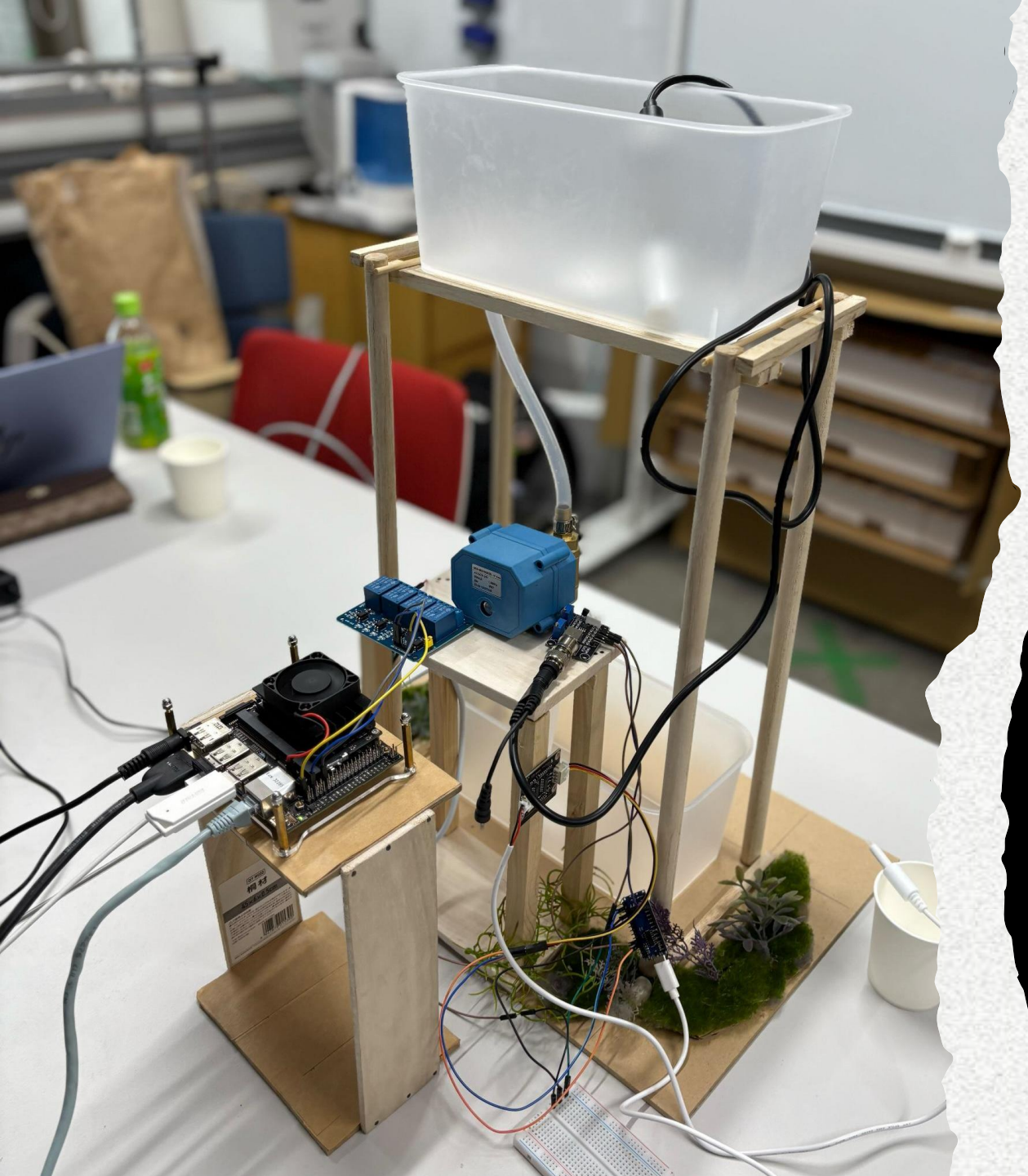


# Anomaly Detection Using Reconstructive Reservoir Computing for Smart Response Water Distribution System

## AI Seminar Final Presentation 2025

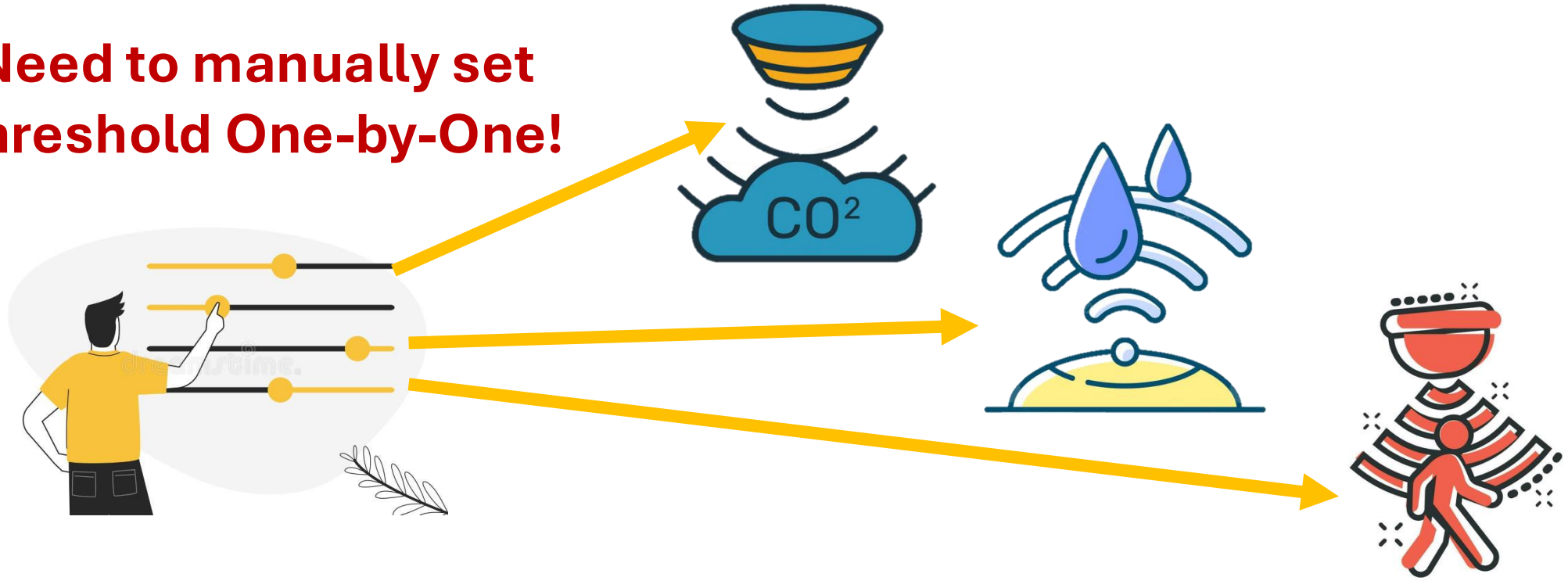
## Group J2 Members


1. Faizah Mappanyompa Rukka 24966078
2. Huynh Truong Tu 24966076
3. Dobhal Umang 24966074
4. Jamloki Prerna 24966070
5. Kanyarak Klaichit 24966067

# Problem with Current Systems

1. ❌ **Manual setup needed** for each sensor and threshold.
2. ❌ **Can only detect extreme values**, not unusual patterns or behavior.
3. ❌ **No automatic response** – tank stays full, so cleaning and recovery take longer.

**Need to manually set threshold One-by-One!**



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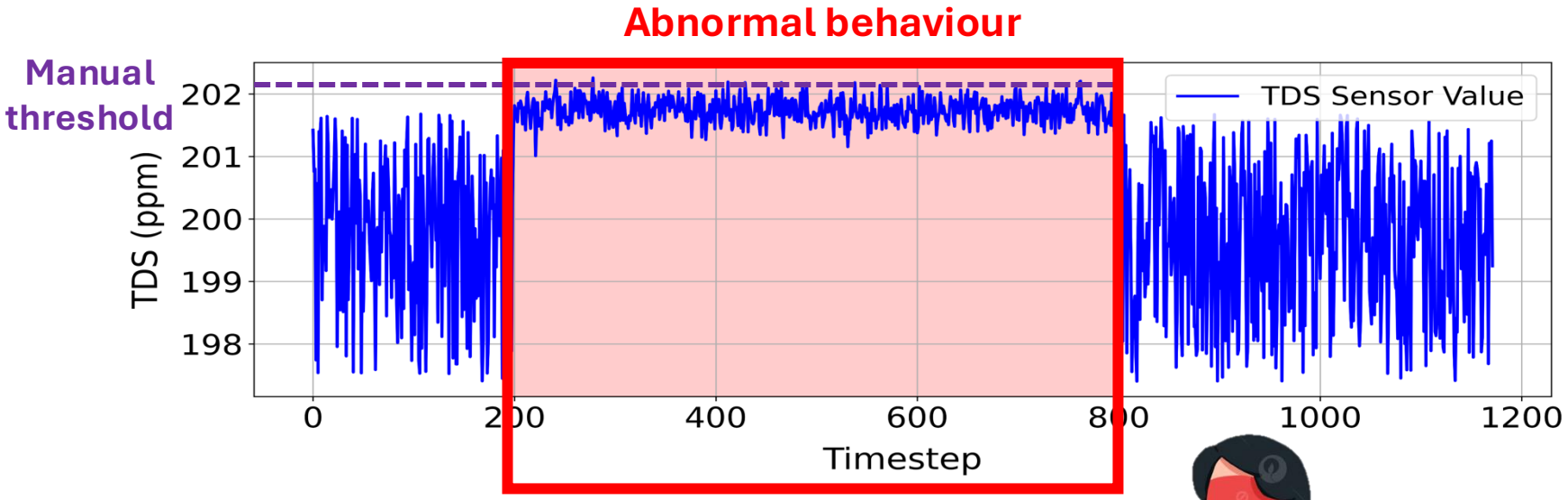
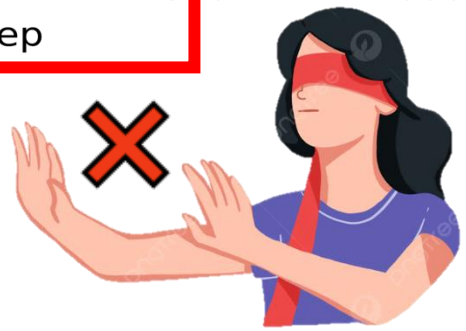



If Manual setting threshold is too high/low



Cannot detect abnormal behaviour

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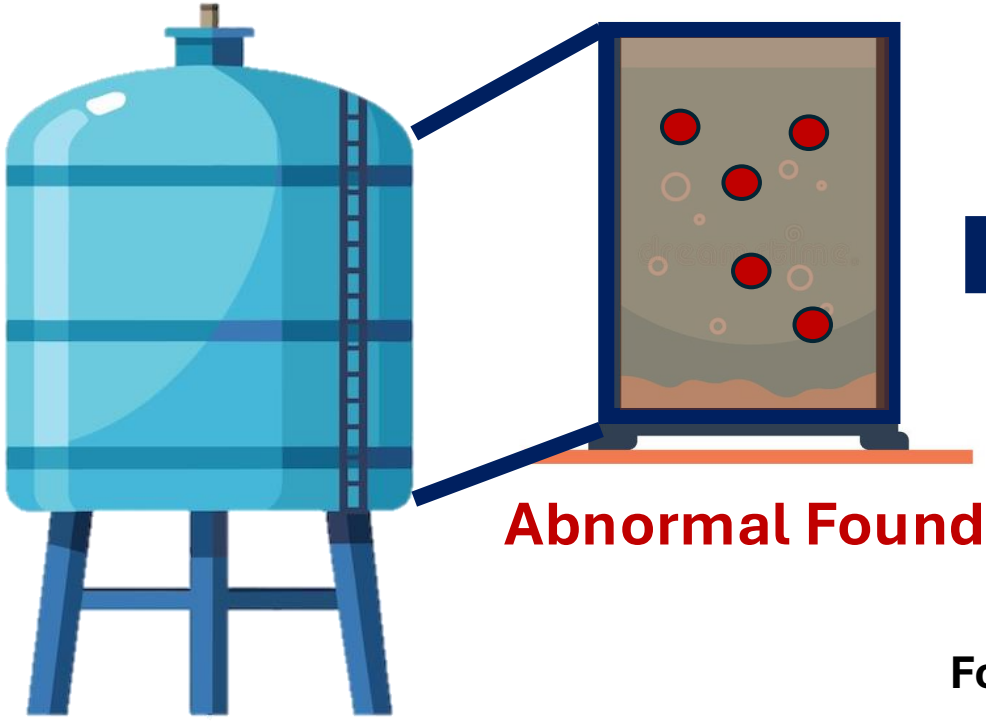


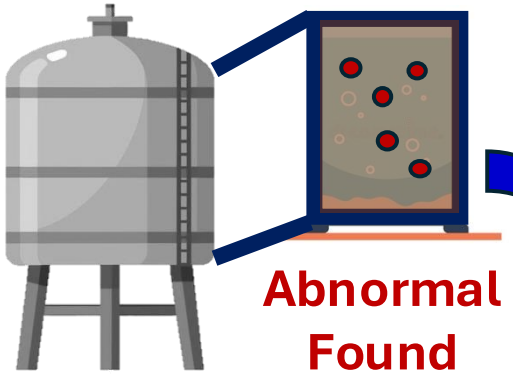
Abnormal Found

Long down time

**Experiment need to be stop**
**For investigation and chemical management**

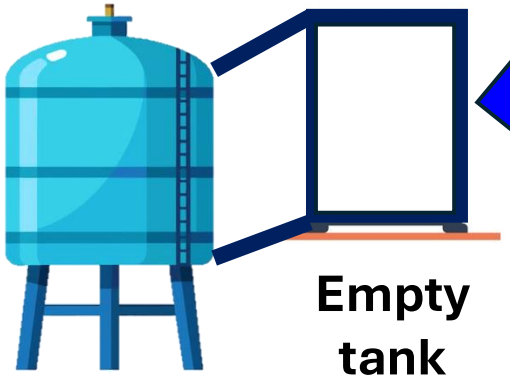
# Proposed Solution

Main Tank



Abnormal Found

2nd Tank



Empty tank

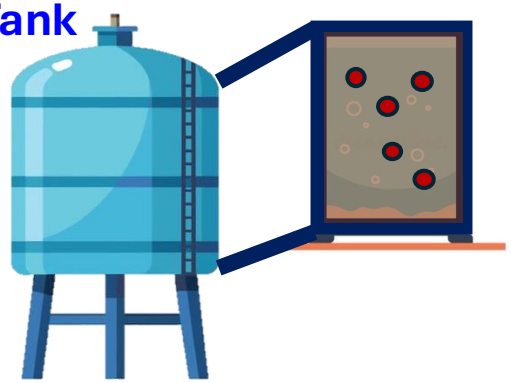
**System**

that can Automatically transfers abnormal waste to a 2nd tank

Main Tank

Ready to be cleaned and reuse quickly

2nd Tank



# Motivation

東京大学
THE UNIVERSITY OF TOKYO

International Joint Conference on Neural Networks (IJCNN) - IEEE 2022

**Proposal of Reconstructive Reservoir Computing to Detect Anomaly in Time-series Signals**

**DOI**: 10.1109/IJCNN55064.2022.9892805

Published January 2024 NOLTA, IEICE

Paper

**Reconstructive reservoir computing for anomaly detection in time-series signals**

**DOI:** 10.1587/nolta.15.183

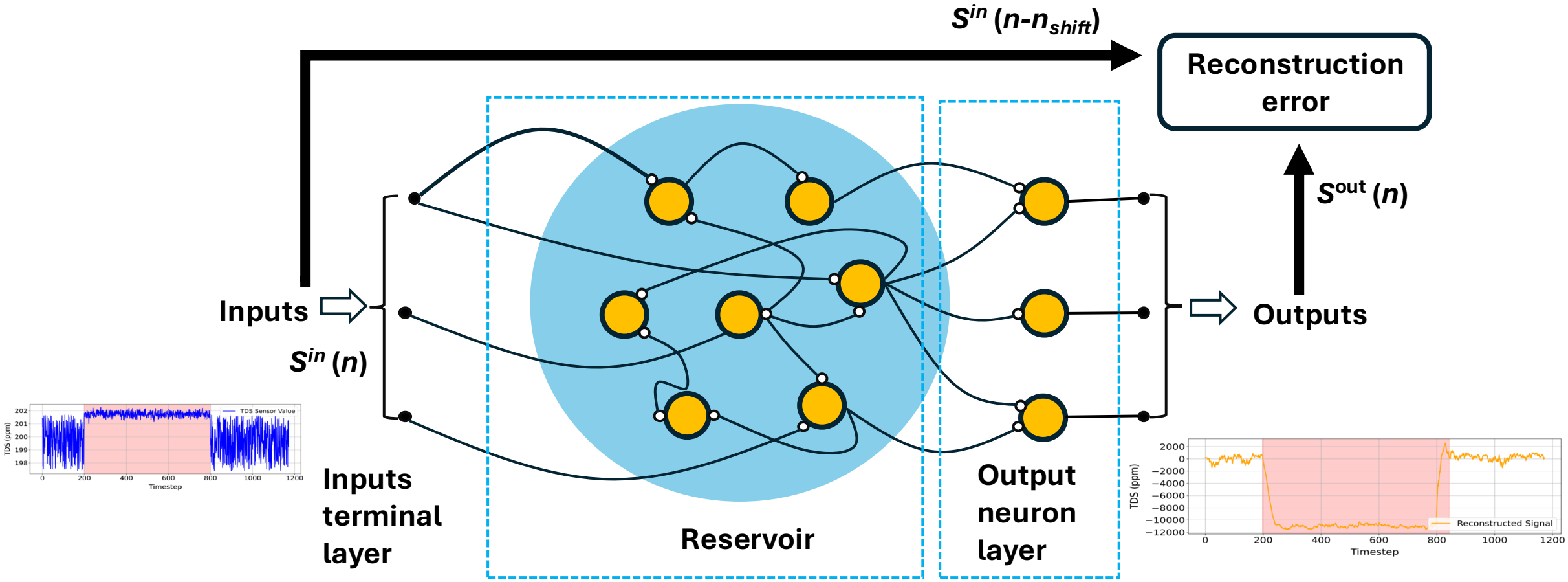
加藤 准也
**Kato Junya**

田中 剛平
**Tanaka Gouhei**

中根 了昌
**Nakane Ryosho**

廣瀬 明
**Hirose Akira**

# ESN (Echo State Network)



# Method

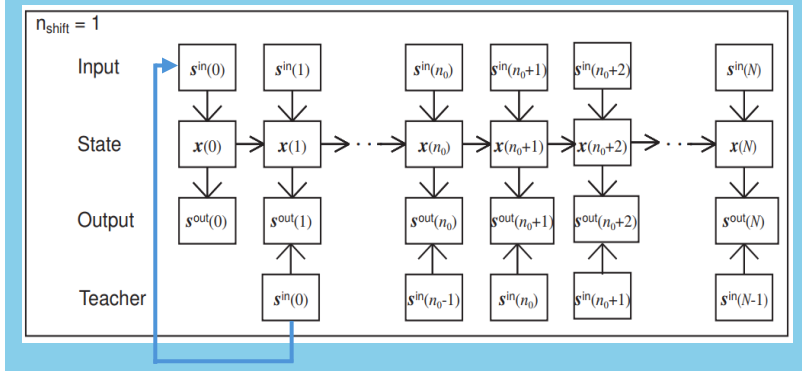
$s^{\text{in}}(n - n_{\text{shift}})$

- $n_{\text{shift}} \geq 0$ : **Reconstruction** (proposed)
- $n_{\text{shift}} \leq -1$ : **Forecasting** (conventional)

**Reconstruction (Proposed)**

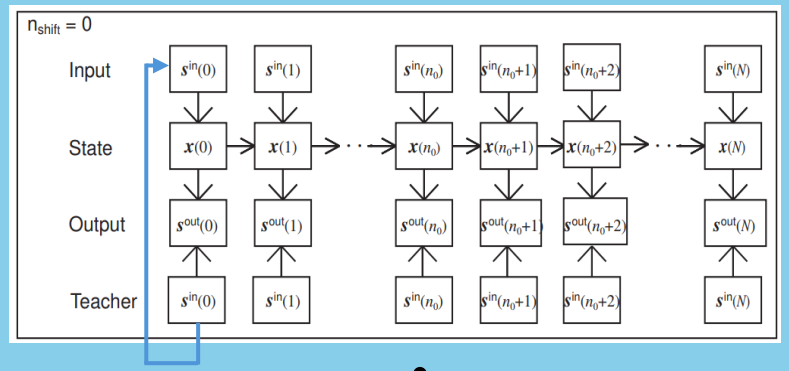
**Forecasting**
Conventional

## Past step



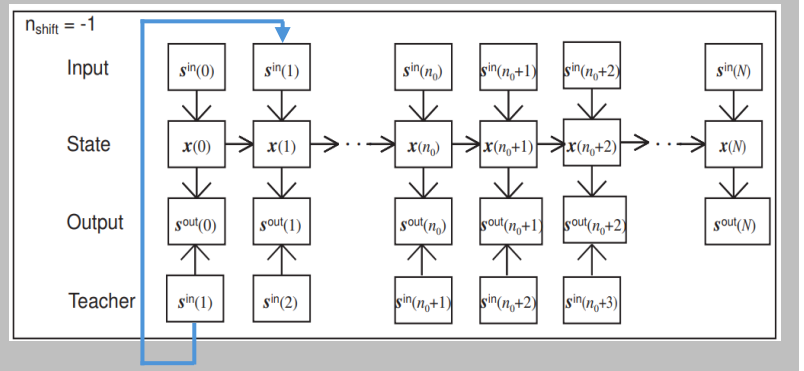

$\mathbf{s}^{\text{in}}(\mathbf{n}-\mathbf{1})$

## Current step




$\mathbf{s}^{\text{in}}(\mathbf{n})$

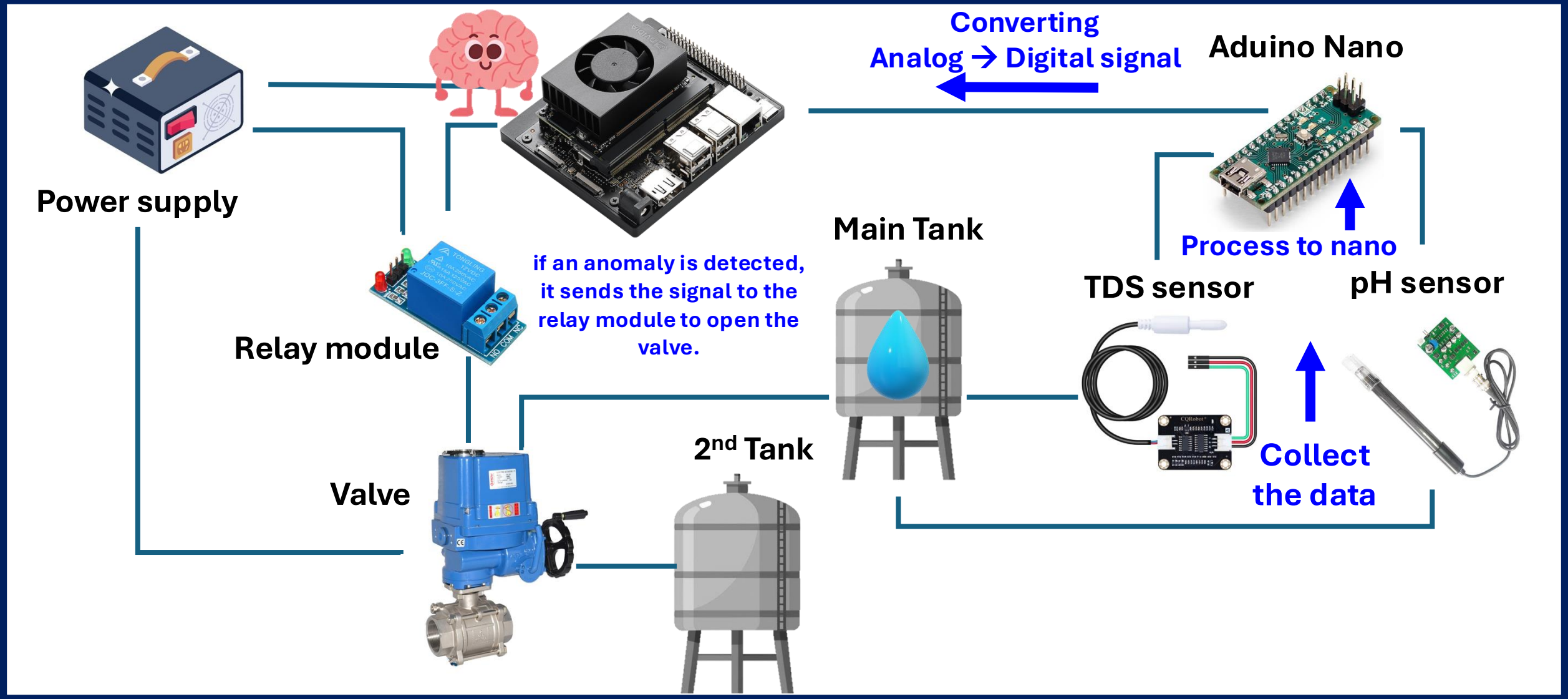
## Future step



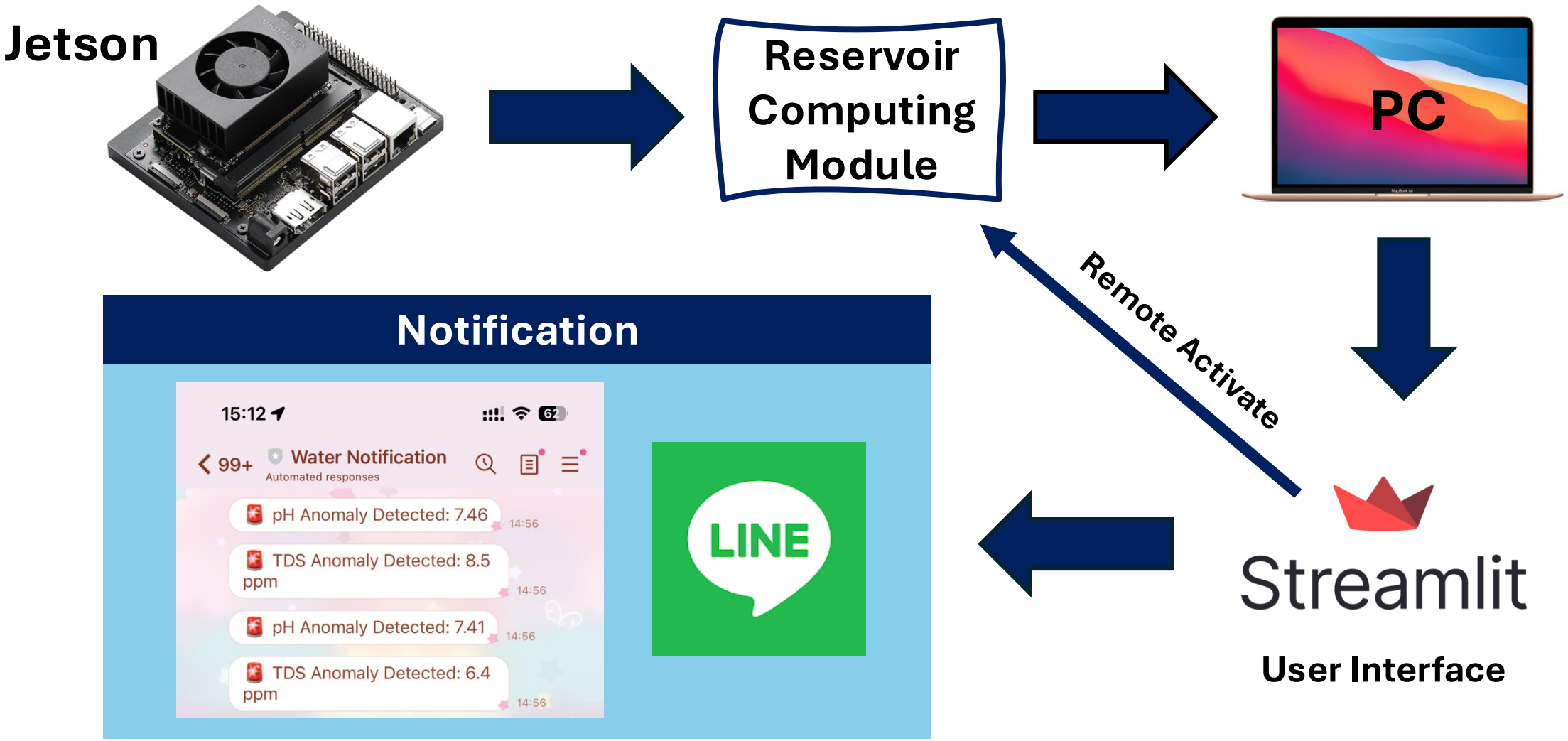

$\mathbf{s}^{\text{in}}(\mathbf{n}+\mathbf{1})$

**Ref:** J. Kato, G. Tanaka, R. Nakane and A. Hirose, *2022 IJCNN*, pp. 1-6, doi: 10.1109/IJCNN55064.2022.9892805

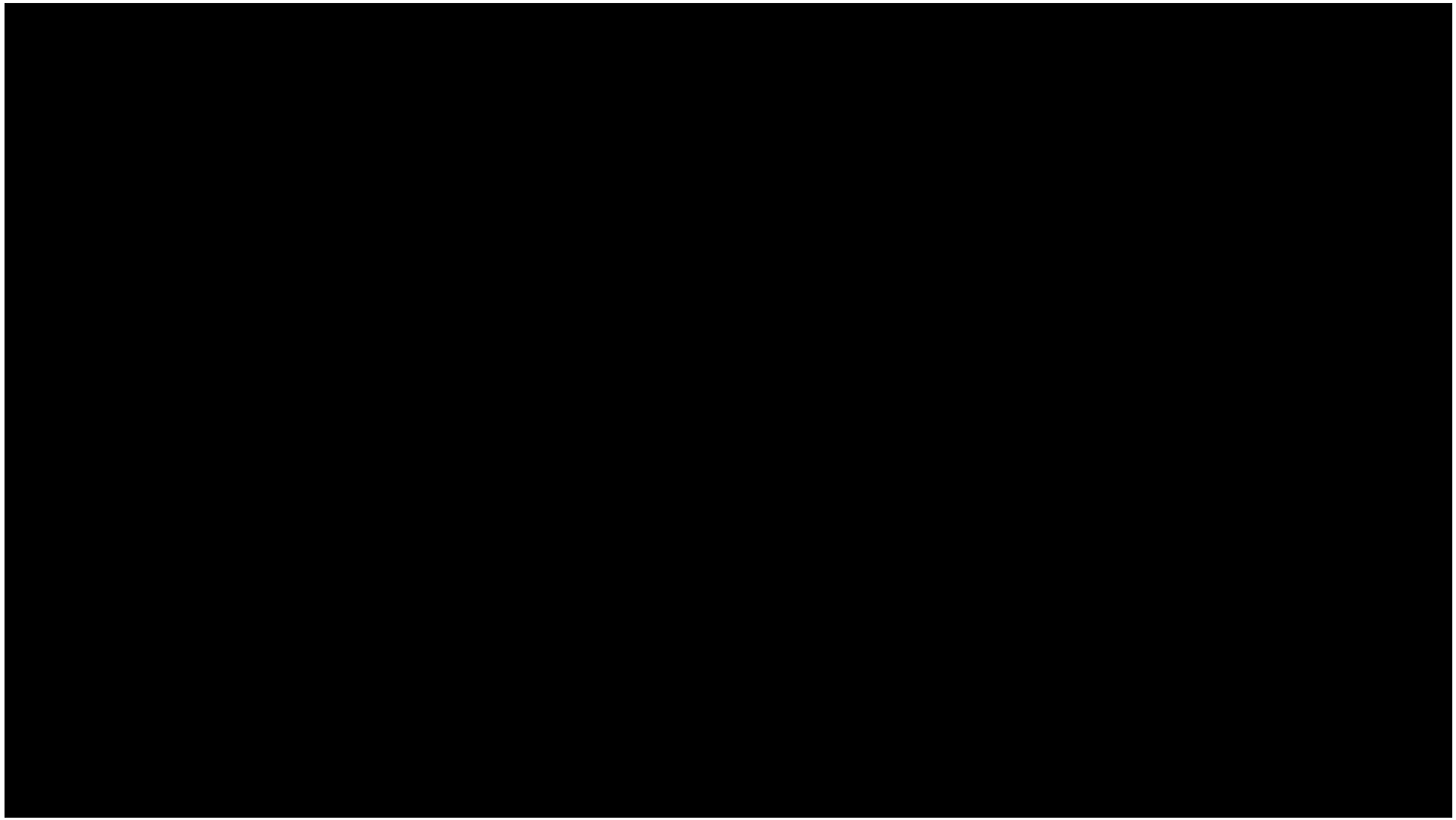
# Diagram


Jetson
Converting
Analog → Digital signal
Aduino Nano
Power supply
Main Tank
Process to nano
if an anomaly is detected, it sends the signal to the relay module to open the valve.
TDS sensor
pH sensor
Relay module
2nd Tank
Valve
Collect the data

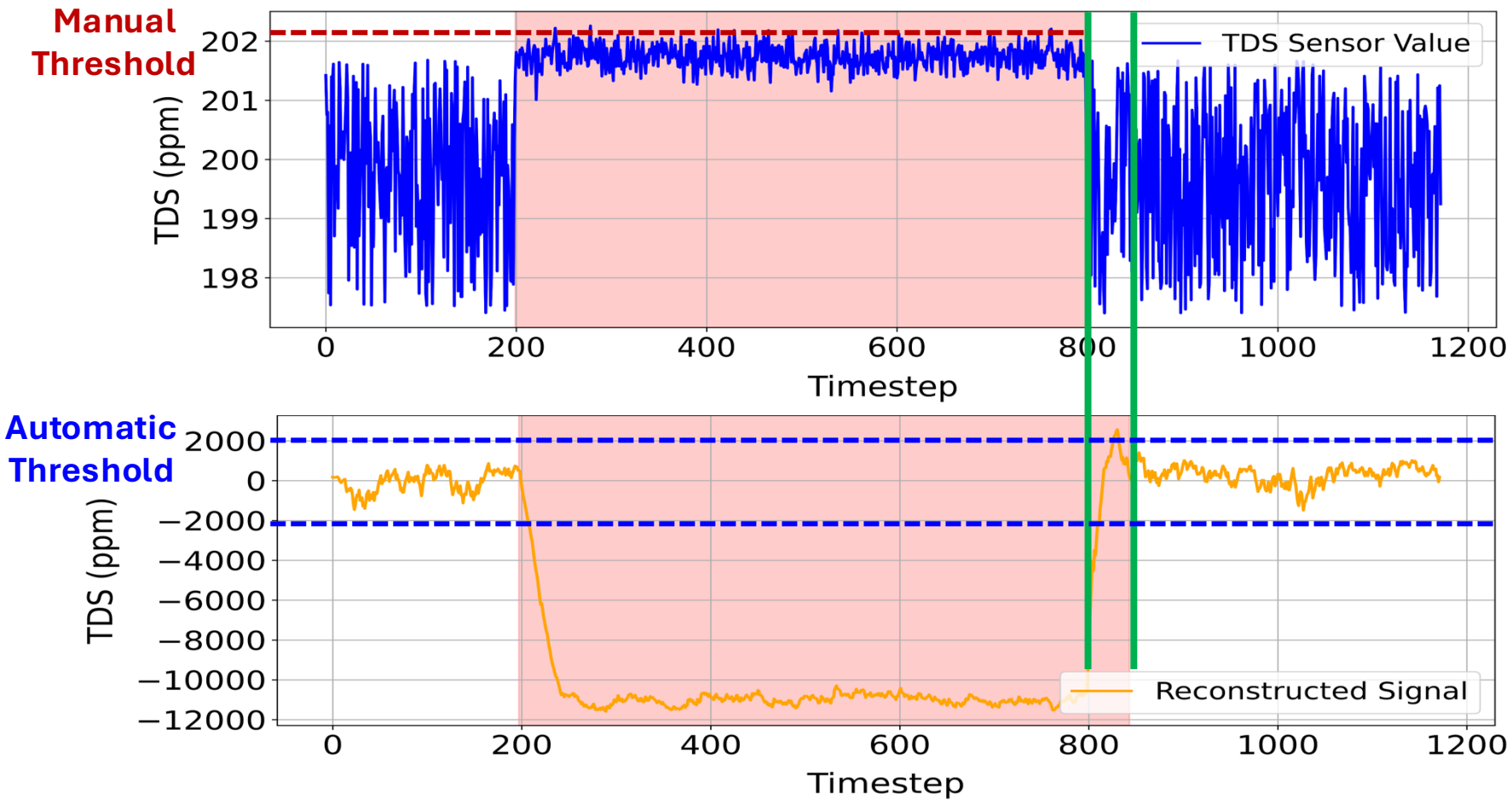
# Notification Diagram



# Demonstration Video



# Result



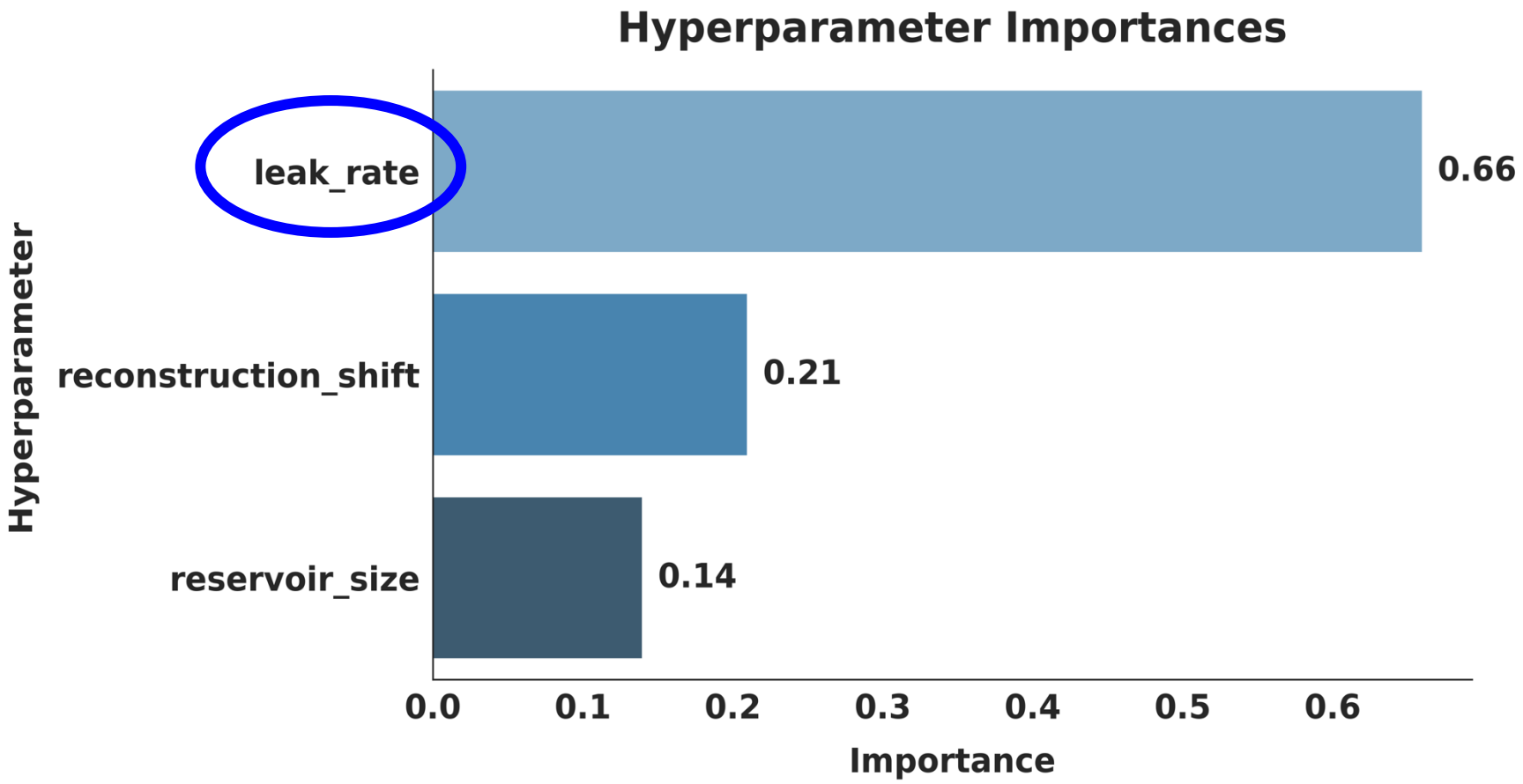

**TDS Raw Data**

**Reconstructive Error**

✅ The system can find abnormal patterns by looking at **signal behaviour and automatically sets a good threshold.**

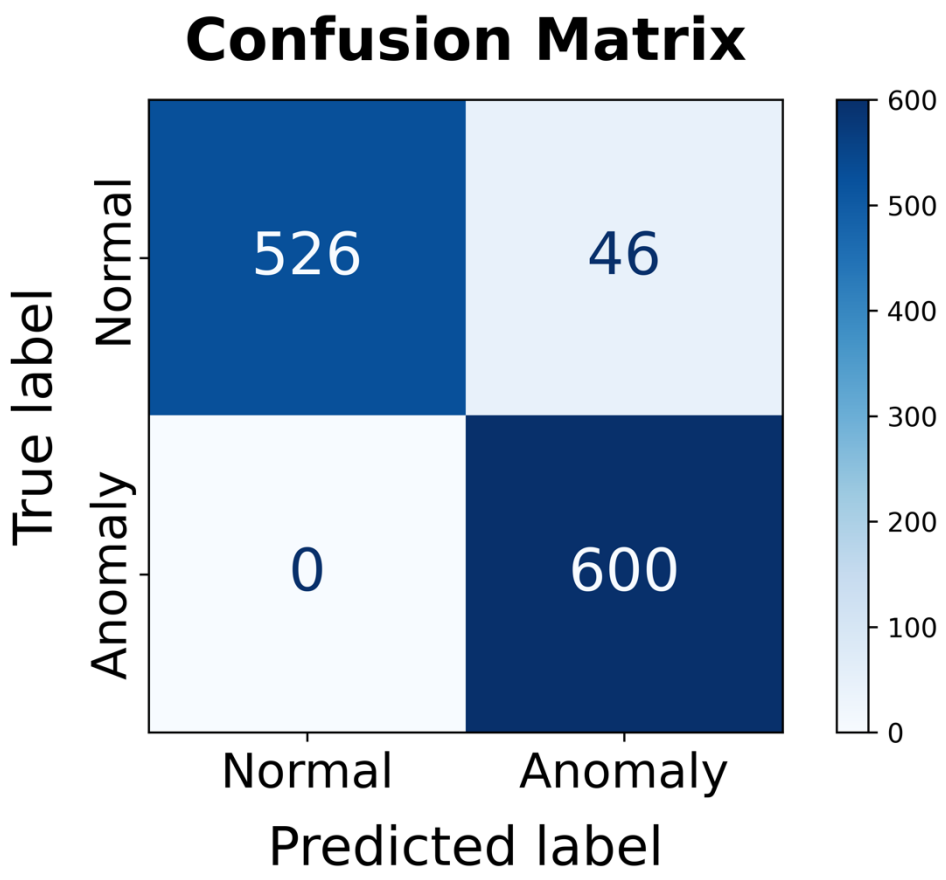
- Unlike **old methods that need manual setting and may miss some cases.**

# Hyperparameter Tunning




✅ Hyperparameter Tuning with **Optuna** (10000 Trials) showed **leak_rate is highest importance.**

# Confusion Matrix




**Best parameters:**

- leak_rate: 0.1
- reservoir_size: 300
- reconstruction_shift: 28

**Metrics:**

**Accuracy = 95.97%**

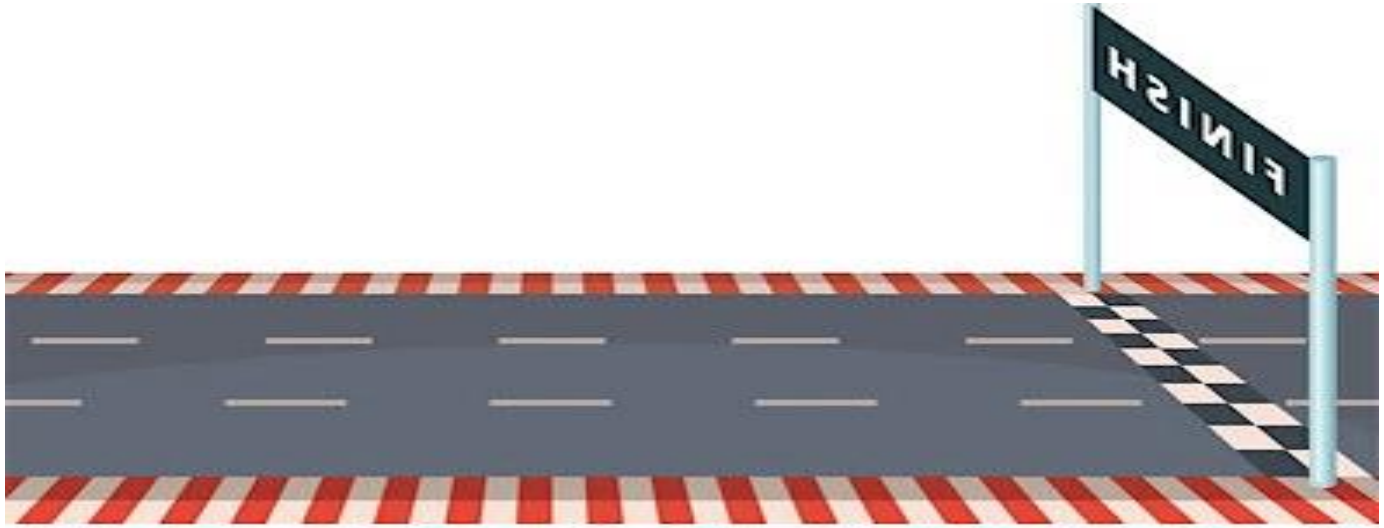
**Normal accuracy = 91.95%**

✅ **Anomaly accuracy = 100%**

✅ **F1 score for Anomaly = 96.33%**

# Conclusion

1. ✅ **Uses behaviour-based analysis** during setup to calculate **efficient fixed thresholds**
2. ✅ In real use, applies those **pre-calculated thresholds** for fast detection – no need for dynamic learning
3. ✅ **Automatically transfers waste to a second tank**, so the main tank can be cleaned and reused quickly
4. ✅ Hyperparameter Tuning with Optuna shows **leak rate is the most important parameter**.



# Thank you
# Q&A